# NUMBER SERIES
# &
# LETTER SERIES

# NUMBER SERIES - LETTER SERIES


– by YASH JAIN

## Solve This Number Puzzle

Find the Next Number in this series:

5, 7, 12, 19, 31, 50, ?


BRUTE FORCE

**Que:** Find the missing number in the series: 3 5 __ 21 43 85

A) 10 B) 12 C) 14 D) 11

Que: Find the missing number in the series: 3 5 __ 21 43 85

A) 10 B) 12 C) 14 D) 11

$3 \longrightarrow 5$

$3 \times p \pm m = 5$

$21 \longrightarrow 43$

$21 \times p \pm m = 43$

$p$

$m$

$\pm$

Que: Find the missing number in the series: 3 5 11 21 43 85

A) 10 B) 12 C) 14 ~~D)~~ 11

1 2 3 4

− + − +

$p = 2$ −

$m = 1$

$43 \to 85$

$43 \times p \pm m$

$(43 \times 2) - 1$

$86 - 1 = 85$

$\times 2 \pm 1$

$p = 2$ +

$m = 1$

$p = 2$ −

$m = 1$

$3 \to 5$

$3 \times p \pm m = 5$

$(3 \times 2) - 1 = 5$

$5 \to ?$

$5 \times p \pm m$

$(5 \times 2) + 1 = 11$

$21 \to 43$

$21 \times p \pm m = 43$

$(21 \times 2) + 1 = 43$

$11 \to 21$

$(11 \times p) - m$

$(11 \times 2) - 1 = 21$

Que: 4, 11, 19, 41, ?, 161

79

A) 62
B) 108
C) 79 ✓
D) 90
E) 85

$(4 \times 2) + 3 = 11$

$(11 \times 2) - 3 = 19$

$(19 \times 2) + 3 = 41$

$(41 \times 2) - 3 = 79$

$\underbrace{(79 \times 2)}_{158} + 3 = 161$

p=2 (+)
m=3

p=2 (−)
m=3

p=2 (+)
m=3

2

Que: 11, 6, 5, 9, 16, ?

A) 66.5
B) 78.5
C) 89.5
D) 42.5
E) 31.5

$p = 2.5$ $\oplus$
$m = 2.5$

$(6 \times 0.5)$
$3 + 0.5$
$3 - 0.5$

$11 \longrightarrow 6$

$(11 \times 0.5) + 0.5 = 6$

$5.5$

$6 \longrightarrow 5$

$(6 \times 1) - 1 = 5$

$5 \longrightarrow 9$

$(5 \times 1.5) + 1.5 = 9$

$9 \longrightarrow 16$

$(9 \times 2) - 2 = 16$

$16 \longrightarrow$ __

$(16 \times 2.5) + 2.5$

$\underline{42.5}$

$\begin{cases} p = 0.5 \\ m = 0.5 \end{cases}$ $\oplus$ ✓

$\begin{cases} p = 1 \\ m = 1 \end{cases}$ $\ominus$ ✓

$\begin{cases} p = 1.5 \\ m = 1.5 \end{cases}$ $\oplus$ ✓

$p = 2$ $\ominus$
$m = 2$

Que: 5, 7, 11, 37, 143, ?

A) 733
B) 721
C) 764
D) 750
E) 784

Que: 5, 7, 11, 37, 143, ?

A) 733
B) 721 ✓
C) 764
D) 750
E) 784

$5 \longrightarrow 7$

✓ $(5 \times 1) + 2$

$7 \longrightarrow 11$

$(7 \times 2) - 3 = 11$

$11 \longrightarrow 37$

$(11 \times 3) + 4 = 37$

$$\begin{array}{r} \overset{2}{37} \\ \times\ 4 \\ \hline 148 \end{array}$$

$37 \longrightarrow 143$

$(37 \times 4) - 5$

$143 \longrightarrow ?$

$(143 \times 5) + 6$

$= 715 + 6$

$= 721$

p = 1, m = 2 (+)

p = 2, m = 3 (−)

p = 3, m = 4 (+)

p = 4, m = 5 (−)

p = 5, m = 6 (+)